

# Performance Comparison of 10 Pre-trained Models on a CIFAR-100 (for 20 classes)

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## Abstract

This report details an experiment to evaluate the performance of various pre-trained deep learning models on a filtered 20-class subset of the CIFAR-100 dataset. Models including VGG16, MobileNet, DenseNet variants, and ResNet variants were adapted using transfer learning. Key challenges included memory constraints when attempting to resize input images to the models' default input sizes, necessitating training on the original 32x32 CIFAR-100 images. Performance varied significantly, with **VGG16** achieving the highest accuracy (0.6305), followed by **DenseNet169** (0.6075). The limited dataset size and the small number of parameters fine-tuned for some models impacted overall performance.

## 1 Introduction

Our goal is to test the performance of various pre-trained models on the CIFAR-100 dataset, specifically focusing on a 20-class subset. We will preprocess the data for these 20 classes from the CIFAR-100 dataset and then test the performance of different pre-trained models including **VGG16**, **MobileNet**, **DenseNet121**, **DenseNet169**, **DenseNet201**, **ResNet101**, **ResNet152**, **ResNet50**, **VGG19**, and **MobileNetV2** on this filtered dataset.

## 2 Dataset Preparation

At first, we load the CIFAR-100 dataset and filter it to include only the first 20 classes. Then we normalize the images such that pixel values are between 0 and 1, which will help in faster convergence during training and get better performance from the models. Then we also one-hot encode the labels for this 20-class subset.

### Problems Faced During Data Preparation

Here, the default input size of many pre-trained models is 224x224, while CIFAR-100 images are 32x32. So, I tried to resize the images to 224x224, to get better performance from the models. But when I tried to resize the images, I faced a problem with memory. My PC was crashing while resizing the images. I also tried to resize the images in a lab PC where I have 16GB GPU memory, but I faced the same problem. I also tried in Google Colab, but I faced the same problem due to the memory limit of Google Colab. So, I had to train the models on the original CIFAR-100 images without resizing them. This affected the performance of the models, and the accuracy of the models is not as good as I expected.

## 3 Training

We will use the pre-trained models from Keras and freeze their layers to retain the learned features. We will then add a custom head for classification and train it on the filtered dataset. During training, we also unfreeze the last few layers of the pre-trained model to fine-tune it for our specific task. Then we will evaluate the model's performance on the test set.

## 4 Evaluation

We will evaluate the performance of each model using accuracy as the primary metric. We will also visualize the training and validation accuracy and loss over epochs to understand how well the model is learning.

## 5 Results and Discussion

### VGG16

**VGG16** achieved an accuracy of 0.6305 on the test set after training. This indicates that the model was able to learn the features of the 20-class subset moderately well. One of the reasons is that **VGG16** is a deep model with many layers, and it was trained on a large dataset (ImageNet), meaning it is not specifically tailored for CIFAR-100. Another reason is that the **VGG16** model's default input size is 224x224, while CIFAR-100 images are 32x32, which may lead to lower performance due to the very small input size of CIFAR-100 images. An important reason is that **VGG16** has a large number of parameters (approximately 14,851,156), but we only trained a small portion of those parameters (4,856,084), which may not be sufficient to learn the specific features of the 20-class subset in CIFAR-100. On the other hand, we do not have sufficient data to train all the parameters of **VGG16**, which may also lead to lower performance.

### MobileNet

**MobileNet** achieved an accuracy of 0.2765 on the test set after training. This indicates that the model's performance is not as good as **VGG16**. The main reason for this is that **MobileNet** is designed to be lightweight and efficient, which may lead to a trade-off in performance compared to deeper models like **VGG16**. Additionally, this model is also not specifically tailored for CIFAR-100, and the input size is also 224x224 while CIFAR-100 images are 32x32, which may lead to low performance. The number of parameters in **MobileNet** is also less than **VGG16** (3,496,404), but we only trained a very small portion of those parameters (1,318,164) during training (because we do not have a sufficient amount of data to train all the parameters), which may not be sufficient to learn the specific features of the 20-class subset in CIFAR-100.

### DenseNet121

**DenseNet121** achieved an accuracy of 0.5780 on the test set after training. This indicates that the model's performance is better than **MobileNet** but not as good as **VGG16**. **DenseNet121** is a deeper model with more layers, which allows it to learn more complex features. The total number of parameters in **DenseNet121** is 7,305,044, and we trained a very small portion of those parameters (433,684) during training.

### DenseNet169

**DenseNet169** achieved an accuracy of 0.6075. This indicates that the model's performance is better than **DenseNet121**. This is because **DenseNet169** is a deeper model with more layers, which allows it to learn more complex features than **DenseNet121**. The total number of parameters in **DenseNet169** is 13,074,260, and we also trained a very small portion of those parameters (680,724) during training.

### DenseNet201

**DenseNet201** achieved an accuracy of 0.6062. This indicates that the model's performance is nearly the same as **DenseNet169**. Though **DenseNet201** is a deeper model with more layers than **DenseNet169**, the performance is not significantly better because we only trained a very small number of parameters (537,620) during training, which is much less than the number of parameters we trained in **DenseNet169**. Considering this, we can say that **DenseNet201** might be considered better than **DenseNet169** in this context because it gives nearly the same performance with fewer trained parameters.

## ResNet101

**ResNet101** achieved an accuracy of 0.2468, a very low performance on the test set after training. This indicates that the model is poorly able to learn the features of the 20-class subset. The main reason for this is that **ResNet101** has a very large number of parameters (43,187,860), and we only trained a very small number of parameters (533,780) due to the insufficient amount of data. Again, the input size is also 224x224, but CIFAR-100 images are 32x32. Another reason is that **ResNet101** is trained on ImageNet, which is a large dataset with many classes, and it may not be able to learn the specific features of the 20-class subset in CIFAR-100.

## ResNet152

**ResNet152** achieved a very poor performance of 0.0884. This indicates that the model is not able to learn the features of the 20-class subset of CIFAR-100 at all. The main reason for this is that **ResNet152** has a very large number of parameters (58,900,628), and we only trained a very small number of parameters (6,050,068) due to the insufficient amount of data. Again, the input size is also 224x224 but CIFAR-100 images are 32x32. A large number of parameters remaining tuned for ImageNet is a big reason for the poor performance on CIFAR-100.

## ResNet50

**ResNet50** achieved an accuracy of 0.3196. This indicates that the model's performance is better than **ResNet101** and **ResNet152**. Because **ResNet50** is a smaller model with fewer parameters (24,117,396) than **ResNet101** and **ResNet152**. From those parameters, we trained 533,780 during training. This means a comparatively larger proportion of its fine-tunable parameters (relative to its overall size or the other ResNets) might have been adapted to CIFAR-100 than for **ResNet101** and **ResNet152**. Again, comparatively fewer parameters remained tuned for ImageNet than **ResNet101** and **ResNet152**. That's why **ResNet50** performed better than **ResNet101** and **ResNet152** on this CIFAR-100 subset.

## VGG19

**VGG19** achieved an accuracy of 0.5913 on the test set after training. This indicates that the model's performance is slightly better than many of the other models but not as good as **VGG16**. The main reason for this is that **VGG19** is much deeper than **VGG16** with a larger number of parameters (20,160,852). Since we trained a very small number of parameters (4,856,084) during training, a very large number of parameters remained tuned for ImageNet. Since CIFAR-100 is a small dataset with limited similarity to ImageNet, **VGG19** offers moderate performance. However, because many more parameters remained tuned for ImageNet than in **VGG16** (relative to the fine-tuning strategy), **VGG19** performed slightly lower than **VGG16**.

## MobileNetV2

**MobileNetV2** achieved an accuracy of 0.3981. This indicates that though the model's performance is worse than **VGG16**, **DenseNet121**, **DenseNet169**, **DenseNet201**, and **VGG19**, it is better than **MobileNet**. The main reason for this is that **MobileNetV2** is designed to be lightweight and efficient, with 2,591,060 total parameters. From those parameters, we trained 745,236 during training. Since **MobileNetV2** is a more advanced version of **MobileNet**, and the number of parameters trained on CIFAR-100 might be a more effective portion, **MobileNetV2** performed better than **MobileNet**.

## Summary Table of Performance

Table 1: Performance Summary of Pre-trained Models on CIFAR-100 (20 classes)

| Model              | Test Accuracy | Total Parameters | Trainable Parameters |
|--------------------|---------------|------------------|----------------------|
| <b>VGG16</b>       | 0.6305        | 14,851,156       | 4,856,084            |
| <b>MobileNet</b>   | 0.2765        | 3,496,404        | 1,318,164            |
| <b>DenseNet121</b> | 0.5780        | 7,305,044        | 433,684              |
| <b>DenseNet169</b> | 0.6075        | 13,074,260       | 680,724              |
| <b>DenseNet201</b> | 0.6062        | 18,818,900       | 537,620              |
| <b>ResNet101</b>   | 0.2468        | 43,187,860       | 533,780              |
| <b>ResNet152</b>   | 0.0884        | 58,900,628       | 6,050,068            |
| <b>ResNet50</b>    | 0.3196        | 24,117,396       | 533,780              |
| <b>VGG19</b>       | 0.5913        | 20,160,852       | 4,856,084            |
| <b>MobileNetV2</b> | 0.3981        | 2,591,060        | 745,236              |

## 6 Conclusion

The performance of various pre-trained models was tested on a 20-class subset of the CIFAR-100 dataset. The results show that **VGG16** achieved the best performance among the models tested, followed by **DenseNet169** and **DenseNet121**. **MobileNet** and **ResNet** models (particularly **ResNet152** and **ResNet101**) performed poorly on this CIFAR-100 subset. The main reasons for the varied performance include the small size of the CIFAR-100 dataset used for fine-tuning, the small 32x32 input size of the images (compared to the models' typical 224x224 pre-training), and the limited portion of parameters fine-tuned for some models due to the insufficient amount of data for full retraining. The performance of these models could potentially be improved by training on a larger dataset or by employing data augmentation techniques on the CIFAR-100 data.